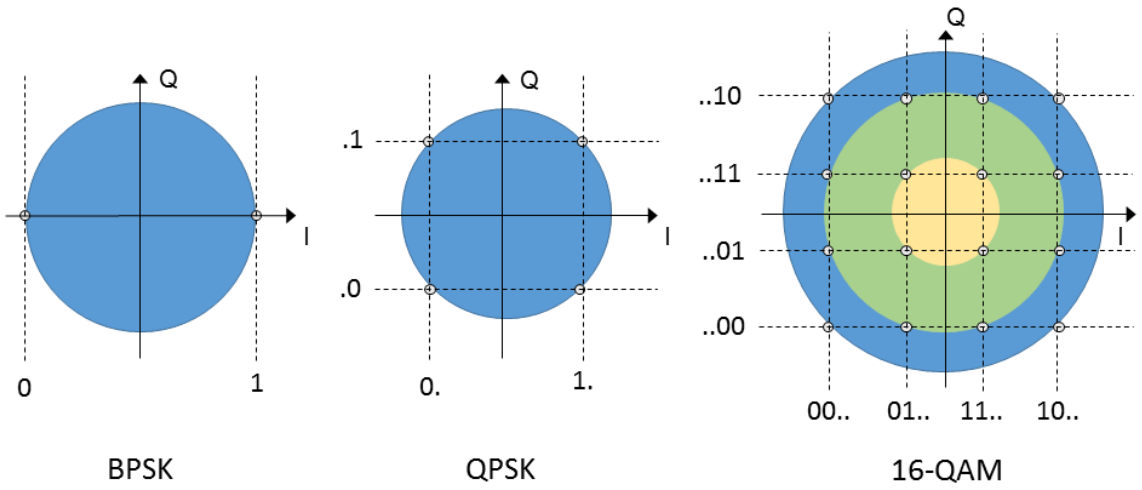
Mobile Internet Technology
SS 2026

Assignment 1

Discussion: 06.05.2026

Exercise 1.1: Signals, Modulation, and Bandwidth

a.) Modulate the ASCII character sequence **MIT26** with the following modulation techniques:



For your solution, assume that each signal has a duration of one oscillation. For representation of the ASCII characters, refer e.g. to wikipedia.

b.) You are designing a wireless communication system that needs to operate with 9.6 kBit/s. You have chosen 16-QAM for modulation. Which bandwidth do you have to assign to your system?

c.) Given is a channel with a bandwidth of 3 MHz. We want to achieve a data rate of 20 MBit/s. What minimum SNR do you need to achieve this data rate? Give its value it dB.

d.) You are able to guarantee a SNR of 20dB for the channel above. Which modulation you can apply on the given channel? Can you achieve the target data rate of 20 Mbit/s? If not, what changes in your system you might do to achieve the target data rate?

Exercise 1.2: Signal Propagation

Assume a connection between a mobile device and a Wi-Fi access point in an 802.11g network on channel 1. The transmit signal power is 100 mW and the receiver requires a signal power of at least 10^{-8} mW. The gains G_i of the antennas are 2.15 dbi. (HINT: find out the center frequency of channel 1 in 802.11g. And: dbi is a **db**-measure referring to an **i**sotropic radiator as a reference – while in your calculations you would have to use the direct relationship, not the db-representation.)

a.) Compute the maximum possible communication distance between the two devices in a free space setup. How does the distance influence the signal that is perceived at the receiver?

b.) By which ratio does the maximum propagation distance decrease assuming a city scenario with $\alpha = 3.5$? Which effects are influencing the signal strength?

c.) How does changing to channel 140 in the 5 GHz band influence the signal propagation distance?

d.) Why are waves with very low frequency and waves with very high frequencies (higher than some hundred GHz) not used for data transmission in computer networks?

Assignment 2

Discussion: 18.05.2026

Exercise 2.1: Multiplexing

For multiplexing, SDM, TDM, FDM, and CDM were presented in the lecture.

a.) CDM systems are often called “much more flexible” than TDM or FDM systems. Reason about this statement: what limits the number of simultaneous users in a TDM/FDM system compared to a CDM system? How does an additional user influence the other users in a cell in FDM, TDM, and CDM systems?

b.) In GSM a combination of TDMA and FDMA is used. Thus we as well have to keep guard spaces as to provide synchronization – we combine the disadvantages of TDM and FDM! Why is such a combination nevertheless done instead of using only FDMA or TDMA?

c.) What are the main benefits of a spread spectrum system?

d.) You want to set up a CDMA-based system for two users. The signal rate is given with r_s = 1 MSignals/s. PSK is used for modulation. Which orthogonal codes you could choose to achieve a data rate of r_1 = 500 kBit/s for user 1 and a data rate of r_2 = 125 kBit/s for user 2? Is it possible to later on add a third user who needs a data rate of 250 kBit/s?

e.) Given are the following chip sequences:

A: (-1 -1 -1 +1 +1 -1 +1 +1)

B: (-1 -1 +1 -1 +1 +1 +1 -1)

C: (-1 +1 -1 +1 +1 +1 -1 -1)

D: (-1 +1 -1 -1 -1 -1 +1 -1)

The stations want to transmit the following bits: A: 1, B: 0, C: 0, D: -. What is the transmitted sequence?

Exercise 2.2: Forward Error Correction with Convolutional Codes

Given is a (3,1,3) convolutional code with

$v_{n1} = u_n$ XOR u_{n-1}, $v_{n2} = u_n$ XOR u_{n-2}, $v_{n3} = u_n$ XOR u_{n-1} XOR u_{n-2}.

a.) Give the shift register diagram and the state diagram for this code.

b.) Assume that the bit sequence `11101` is to be encoded. Give the output of the encoder.

c.) Give the Trellis diagram for the code. What is the free distance of the code?

d.) You are receiving the sequence `11011011011101010101`. Find the most likely bit sequence that was transmitted using the Viterbi algorithm in terminated mode.

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Assignment 3

Discussion: 22.05.2026

Exercise 3.1: CSMA and MACA

Six stations are grouped around a mountain in a chain topology. Each station is able to receive only the two neighbouring stations (station 1 and 6 can also hear each other). Stations perform CSMA, but without collision handling. Consider the following scenarios:

(i) Station 2 currently is sending to station 1. Station 3 wants to address station 4. Can station 3 start transmitting, and should/could it be allowed to do so?

(ii) Station 3 currently is sending to station 2. Station 5 wants to address data to station 4. Can station 5 start transmitting, and should/could it be allowed to do so?

(iii) Stations 1 and 2 are currently sending. Which other stations belief that they can start transmitting, and which ones should/could actually be allowed to do so?

(iv) Stations 1 and 4 are currently sending. Which other stations belief that they can start transmitting, and which ones should/could actually be allowed to do so?

Exercise 3.2: Hidden and Exposed Stations

a.) Consider the problem of hidden and exposed stations. Do you have hidden or exposed stations if Aloha, Slotted Aloha, Reservation Aloha, or MACA is used?

b.) How can MACA still fail in presence of hidden stations?

Exercise 3.3: WLAN and OFDM

a.) In 802.11a, OFDM with 52 subcarriers is used. 4 of these subcarriers are used as pilots. What is the purpose of such a “pilot” and why are there four (and not only a single one)?

b.) 48 OFDM subcarriers are used in 802.11a, and modulation up to 64-QAM is applied. One OFDM signal has a length of 3.2 µs, and between two OFDM signals there is left a guard time of 0.8 µs (for avoiding inter-symbol interference). Which maximum data rate could be achieved? Why is it higher than the specified data rate of 54 MBit/s?

Assignment 4

Discussion: 12.06.2026

Exercise 4.1: WLAN performance

The efficiency of a network heavily depends on its MAC. WLAN is one of the most prominent wireless networks today, but how efficient is medium access? Exemplarily, have a closer look to the standard 802.11b.

Let the traffic in an 802.11b network consist of UDP packets each containing 1500 Bytes of payload. Assume long frame format (see slide III-77). The maximum bit rate of 11 MBit/s can be maintained between all wireless stations and the AP. Estimate the maximum throughput (i.e. payload data rate) a station can achieve under these conditions. *(Note: there are several aspects in the calculation which could be considered differently or depend on configuration of the network. Just give an approximate solution.)*

Exercise 4.2: WLAN Medium Access

Have a closer look at some aspects of the medium access in 802.11.

a.) A station that successfully transmitted a frame and immediately wants to transmit a second frame, does have to choose a random backoff in the next transmission attempt. What is the reason for doing so?

b.) Each frame in 802.11 has to be acknowledged. In Ethernet, no acknowledgements are used. Why are acknowledgements necessary in 802.11, but not in Ethernet?

c.) Assume that an 802.11b station is configured to always use the RTS/CTS handshake before transmitting a frame. This station now wants to transmit a frame of 1000 Byte size. How long does it take for the station to successfully transmit the frame?

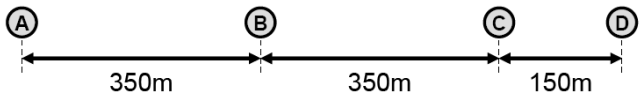
Assume that no other station currently is transmitting or wants to transmit, maximum data rate can be maintained, and no bit errors occur. Additionally, neglect the latency on the wireless link.

d.) 802.11 allows for mobility support – if a station moves between several BSSs of the same ESS, the distribution system always keeps track of where a station currently resides.

Assume that all APs of an ESS are connected with a single switch as distribution system. Sometimes, an AP has to transmit a frame with a faked MAC address to allow the switch to correctly forward the following frames. Why is this necessary?

Exercise 4.3: WLAN in Ad-hoc Mode

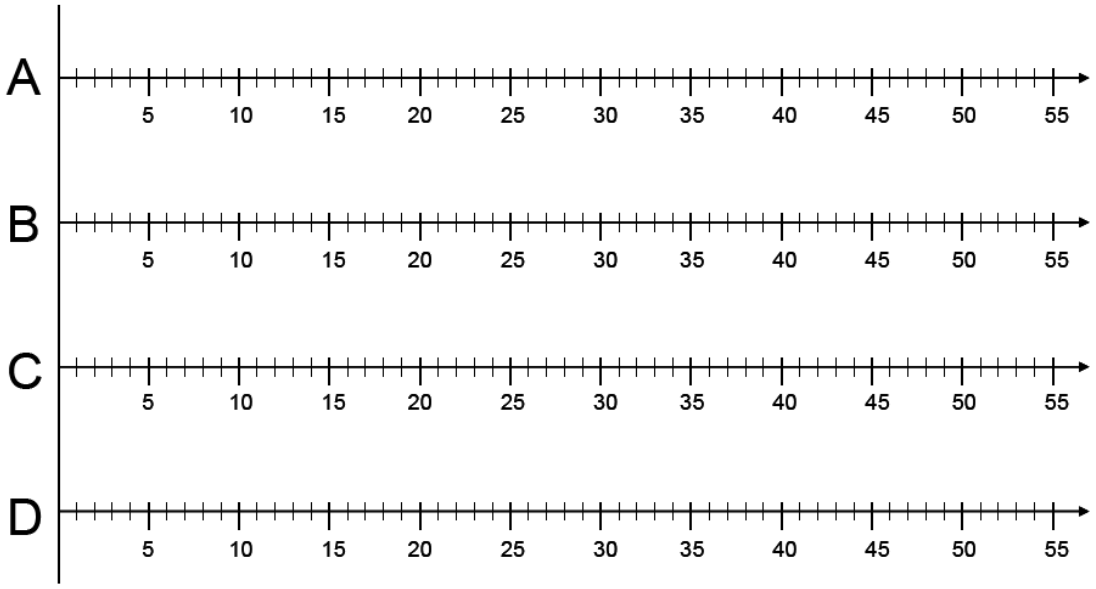
Given is the following ad-hoc network based on WLAN:



The transmission range of all stations is 375m, the interference range is 670m. The stations are sharing the same channel and have sending wishes as shown in the following table:

Sender	Receiver	Time	Sending duration	Chosen backoff
D	**C**	**1**	**2**	**-**
C	**D**	**5**	**4**	**-**
B	**A**	**8**	**6**	**3**
A	**B**	**21**	**6**	**-**
D	**C**	**23**	**5**	**-**
D	**C**	**30**	**4**	**-**
B	**A**	**32**	**4**	**4**
C	**D**	**36**	**3**	**-**
C	**Broadcast**	**45**	**4**	**2**
D	**C**	**46**	**3**	**2**

Assume that the DIFS interval has a duration of 1 time unit. In the following time diagram, sketch the resulting medium use, assuming that unreliable data transfer is used, i.e. no ACKs are sent, no retransmission is done in case of an error.



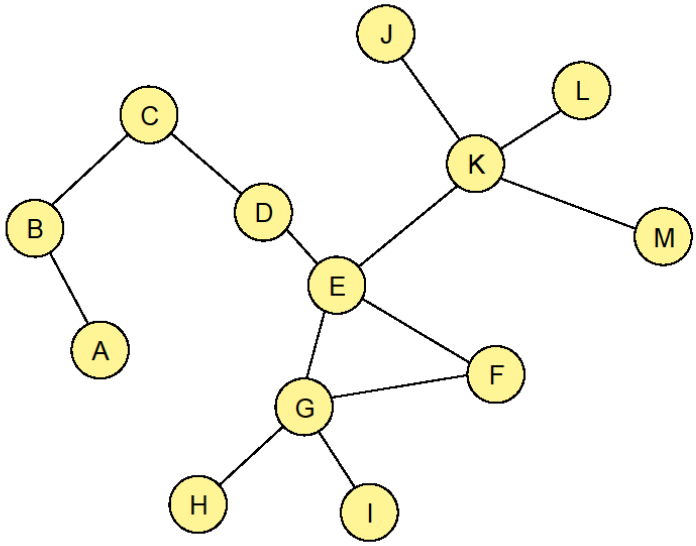
Assignment 5

Discussion: 24.06.2026

Exercise 5.1: Routing in wireless mesh networks

a.) Think of ad-hoc networks with fast moving nodes, e.g., cars in a city. What problems arise even for the routing algorithms adapted to ad-hoc networks? How is the situation changed on highways?

b.) Given is a mesh network which initially has the following topology:



Nodes connected in this picture are in transmission range and can forward messages along the edges. For routing, DSR should be used. The caches of all nodes are empty. Now, *D* wants to send some packets to *H*. Give the sequence of messages exchanged for finding a path. For each message, give sender, receiver, and the list of passed nodes as written in the header by DSR. Assume that the paths are symmetrically.

c.) What happens if the connection between *E* and *G* breaks?

Exercise 5.2: GSM – Cells and Channels

The moon recently has been populated, but till now nobody has installed a telephony system. Thus you decide to install the first GSM system on the moon. You have given the following premises:

- The given frequency band is 880.0 – 882.8 MHz and 935.0 – 937.8 MHz.
- You use a cluster size of 7.
- Only full traffic channels are used.
- Only one TDMA channel per cell is needed for signalling and control.
- There are 12.3 million inhabitants, with constant population density in the whole coverage area.
- Each inhabitant will use the telephony service with an average probability of 1%.

Give an estimation: how many cells are needed in your installation?

Exercise 5.3: GSM System

a.) Given is the following situation: you are subscribed to the GSM network of E-Plus. You are travelling to Kraków; on arrival, you switch on your mobile phone and log in into the Orange GSM network. Nearly at the same time, a friend who is subscribed with T-Mobile, tries to call you. What happens in the following two situations:

(i) the friend tries to call you before you switched on your phone.

(ii) the friend tries to call you after you switched on your mobile phone.

Sketch both scenarios with all involved GSM subsystems.

b.) For data transmission, on average a higher transmit power is used than for voice transmission. Why?

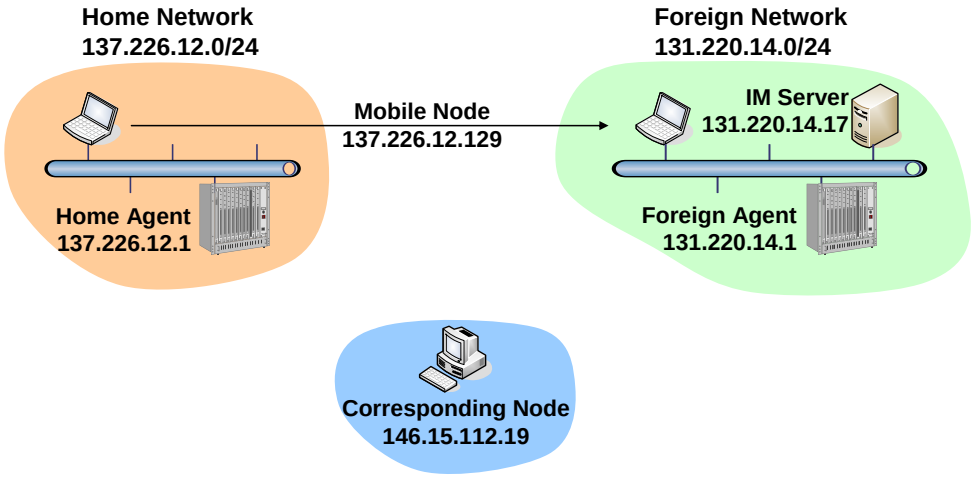
c.) Consider a handover situation. What problems from a user's point of view could occur with a handover if you have an active HSCSD or GPRS connection?

Assignment 6

Discussion: 08.07.2026

Exercise 6.1: Mobile IP Scenario

a.) Have a look at the following scenario:



User *A* at host 146.15.112.19 wants to send a message to user *B* at host 137.226.12.129. For delivery of the message it should be used an instant message server with address 131.220.14.17. User *B* is mobile and currently wants to join the foreign network 131.220.14.0/24.

- How is data in general delivered from user *A* to user *B*? Give all actions performed by the involved components, beginning with the mobile node joining the foreign network, and additionally describe the message flow for delivering data to *B*. Distinguish between the two modes supported by Mobile IP (FA-managed CoA and co-located CoA).
- What is the flow of an instant message from user *A* to user *B* if the foreign agent manages the CoA?

b.) Assume that a mobile node joins a foreign network which uses NAT. Could there be any problems with Mobile IP in presence of NATs? If yes, which problems do arise?

c.) What are main advantages of Mobile IPv6 compared to Mobile IPv4?

Exercise 6.2: Mobile IP, HIP, and xG networks

a.) Compare the main components of Mobile IPv4 with xG networks: is there any concept in an xG network corresponding to Home Agent, Foreign Agent, host's original address, and Care-of-Address? What is similar to these concepts, where are differences? Does xG networks offer functionality not implemented in Mobile IP or vice versa?

b.) Can you find the indirection point of Mobile IP (i.e. the home agent) in the HIP architecture? If yes, where is it? If no, why not?